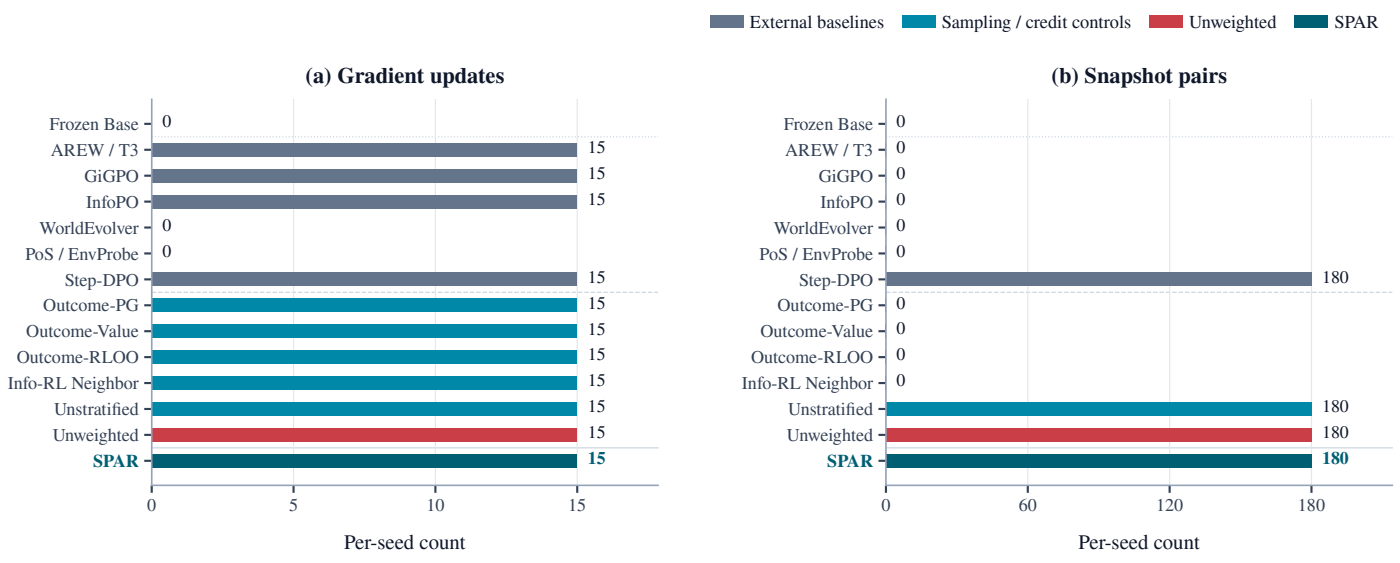
External baselines
Sampling / credit controls
Unweighted
SPAR
(a) Gradient updates
Frozen Base
AREW / T3
GiGPO
InfoPO
WorldEvolver
PoS / EnvProbe
Step-DPO
Outcome-PG
Outcome-Value
Outcome-RLOO
Info-RL Neighbor
Unstratified
Unweighted
SPAR
0
15
15
15
0
0
15
15
15
15
15
15
15
15
0
5
10
15
Per-seed count
(b) Snapshot pairs
Frozen Base
AREW / T3
GiGPO
InfoPO
WorldEvolver
PoS / EnvProbe
Step-DPO
Outcome-PG
Outcome-Value
Outcome-RLOO
Info-RL Neighbor
Unstratified
Unweighted
SPAR
0
0
0
0
0
0
180
0
0
0
0
180
180
180
0
60
120
180
Per-seed count